



Analyzing Speech to Detect Financial Misreporting

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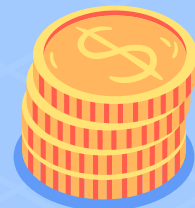


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Motivation

- Financial misreporting erodes investor **trust** and is difficult to detect from numbers or text alone.
- **Research question:** Do CEO's exhibit measurable vocal stress when firms misreport?





Why Non-Verbal Communication Matters



- Beyond words, psychological research shows that emotions such as fear, anxiety, guilt, and shame, often felt by deceivers, are **conveyed through vocal characteristics**.
- Even when CEOs manage content carefully, their voices may still reveal cognitive strain through subtle changes in pitch, pauses, and speech rate.



Research Question

Do CEO's exhibit measurable vocal stress when firms misreport?





Technical Approach



Voice



**Layered Voice
Analysis
(LVA)**



**Cognitive
Dissonance**



Technical Approach



Voice



**Layered Voice
Analysis
(LVA)**



**Cognitive
Dissonance**



Technical Approach



Step 1: Lab Experiment with Students

Check if voice analysis can actually detect cognitive dissonance.



Step 2: Test on Financial Misreporting

See if the same vocal signal predicts real financial misreporting.





Technical Approach



Step 1: Lab Experiment with Students

- Put students into situation where they experience cognitive dissonance
- Approximate the amount of cognitive dissonance with heuristics
- Check if LVA metric predicts cognitive dissonance heuristics (Regression)

Outcome: Voice-based LVA metric significantly predicts behavioral signs of inner conflict



Technical Approach



Voice



**Layered Voice
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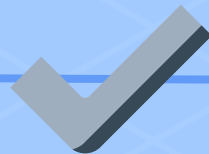
Technical Approach



Voice



**Layered Voice
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**Cognitive
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Technical Approach



Step 2: Test on Financial Misreporting



Voice



Layered Voice
Analysis
(LVA)



Cognitive
Dissonance



Technical Approach



Step 2: Test on Financial Misreporting

- Collect real CEO speech from earnings-call recordings
- Measure vocal dissonance with the validated LVA metric
- Link vocal dissonance to future financial restatements

Outcome: CEOs with higher vocal dissonance were significantly (11%) more likely to issue fraudulent restatements

Key Findings

TABLE 4

Logistic Regression Estimation of the Association Between Proxies for Financial Misreporting, Vocal Dissonance Cues, and Financial Statement–Based Predictors

Panel A: Results without including control variables

Variable ^a	Predicted Sign	(A) <i>ADV.RES</i>	(B) <i>ERROR.RES</i>	(C) <i>IRREG.RES</i>	(D) $-1 \times \text{RESCAR}$	(E) <i>IRREG.RES</i>	(F) $-1 \times \text{RESCAR}$	(G) <i>IRREG.RES</i>	(H) $-1 \times \text{RESCAR}$
Intercept	(?)	−3.055*** (0.332)	−3.221*** (0.398)	−4.601*** (0.558)	−0.024 (0.050)	−4.197*** (0.388)	0.071 (0.052)	−5.044*** (0.657)	−0.042 (0.048)
Vocal Dissonance Marker <i>COGDIS</i>	(+)	2.574* (1.607)	0.853 (1.982)	5.068** (2.464)	0.622** (0.331)				
Financial Statement–Based Predictor <i>FSCORE</i>	(+)					0.418*** (0.173)	0.019 (0.030)		
<i>ACCT-RISK</i>	(+)							0.026*** (0.010)	0.003** (0.001)
Model Type		Logit	Logit	Logit	OLS	Logit	OLS	Logit	OLS
Pseudo R^2 or Adj R^2		0.005	0.001	0.017	0.057	0.017	0.007	0.049	0.113
# of observations ^b		1,572	1,572	1,572	111	1,572	111	1,572	111
AUC ^c		0.552	0.514	0.610	N/A	0.599	N/A	0.690	N/A
Z-stat for Test AUC = 0.500 ^c		1.745	0.379	2.314	N/A	2.086	N/A	5.074	N/A
p Value for Test AUC = 0.500 ^c		0.081	0.705	0.021	N/A	0.037	N/A	<0.001	N/A

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Key Findings

- The LVA vocal metric validly reflects cognitive dissonance
- CEOs with higher vocal dissonance are 11% more likely to misreport





Social & Ethical Implications



Positive Implications

Could enhance/promote transparency in our economic institutions and aid in early fraud investigations.



Firms and Incentivization for Gaming

CEOs might alter their speech strategically if aware of monitoring.



Ethical and Reputational Risk

Misuse may irreparably harm reputations, amplify bias, or erode privacy.



Limitations & Future Work



Correlation $\neq$ Causation

Vocal stress may reflect performance pressure or context, not deception. Controls mitigate but cannot eliminate this risk.



From Research to Real-World Use

Assess real-world impact before deployment in governance or regulation.



Generalizability

Study was limited to US/English samples; consider using broader, **multilingual datasets**.



Black Box Algorithm

The Layered Voice Analysis (LVA) system is proprietary. Researchers can't see what it actually measures.

Discussion Questions



1. Should voice analysis be used to detect deception or misconduct in real-world settings, with and without consent and transparency?
2. Do you think that voice-based fraud detection could be more useful to regulators, companies, investors, the public...or, at the end of the day, no one?



Appendix

TABLE 4 — *Continued*

Panel B: Results after including control variables

Variable ^a	Predicted Sign	(A) <i>IRREG_RES</i>	(B) <i>IRREG_RES</i>	(C) <i>IRREG_RES</i>	(D) <i>IRREG_RES</i>	(E) <i>IRREG_RES</i>	(F) <i>IRREG_RES</i>
CEO Characteristics							
<i>AGE</i>	(?)				−0.009 (0.046)	0.006 (0.046)	0.003 (0.045)
<i>TENURE</i>	(?)				0.011 (0.036)	0.012 (0.039)	0.013 (0.039)
Monitoring							
<i>ALLSTAR_QUES</i>	(+/-)				0.583 (0.530)	0.846 (0.531)	0.920* (0.524)
Model Type		Logit	Logit	Logit	Logit	Logit	Logit
Pseudo R^2		0.034	0.064	0.075	0.054	0.115	0.128
# of Observations ^b		1,572	1,572	1,572	1,572	1,572	1,572
AUC ^c		0.653	0.705	0.723	0.714	0.780	0.787
Z-stat for Test AUC = 0.500 ^c		3.196	4.639	5.250	6.495	9.173	8.392
<i>p</i> Value for Test AUC = 0.500 ^c		0.001	< 0.001	0.002	< 0.001	< 0.001	< 0.001